End-TermExamination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name: B.Tech. (CSE-AI), Semester: 5th Semester
(Nov-Dec, 2022)

Subject Code: BAI-303 **Subject: Cyber Security** **Maximum Marks :60**

Time :3 Hours

Note: Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1 Conceptually analyze the following with suitable examples: (2.5*8=20)

(a) Session Hijacking
(b) Advanced Persistent Threat
(c) Shoulder Surfing
(d) SSH Tunnel
(e) Email Spoofing
(f) Multifactor Authentication
(g) Spear Phishing
(h) Obfuscation

UNIT-I

Q2 Conceptually discuss the functioning of the Rootkits and its different types. Suggest suitable defensive mechanisms against Rootkits. (10)

Q3 Discuss the significance of Backdoor in Cyber Security Realm. How do Backdoor Attacks work in real world scenario? Explain the functioning with a suitable example. (10)

UNIT-II

Q4 Analytically discuss the email security threats viz. Deceptive Phishing, Whaling and Pharming. Discuss the email security best practices being adopted in Industry 4.0. (10)

Q5 Elucidate the significance of Reconnaissance in the Cyber Attack Process. Analytically explain related techniques viz. email harvesting, port scanning and vulnerability scanning. (10)

UNIT-III

Q6 Discuss the significance of secure Virtual Storage Space on the Internet. Explain concepts of tokenization, obfuscation and penetration testing in the context of Cloud Security. (10)

Q7 Conceptualize and highlight the significance of SSID (Service Set Identifier) in Wi-Fi Security. Suggest suitable preventive measures to ensure robust and secure functioning of Wi-Fi. (10)

UNIT-IV

Q8 Categorically differentiate between Symmetric and Asymmetric Key Encryption algorithms. Elucidate with suitable mathematical examples. (10)

Q9 Conceptually analyze the significance of Message Authentication Codes (MAC). Discuss the important properties and limitations of MAC. (10)